

Nils Degardin
Midterm 2

Algebra-Based Physics

Task 4: $F = qv \times B \rightarrow v \times B = F$

1)

a) given: $v = (0, +1, 0)$

$F = (-1, 0, 0)$

$(0, 1, 0) \times (B_x, B_y, B_z) = (-1, 0, 0)$

$\Rightarrow v_y B_z = -1 \Rightarrow B_z = -1$ B points into page $(-z)$

b) given: $v = (0, +1, 0)$

$F = (0, 0, -1)$

$(0, 1, 0) \times (B_x, B_y, B_z) = (0, 0, -1)$

$\Rightarrow -v_y B_x = -1 \Rightarrow B_x = +1$ B points to the right $(+x)$

c) given: $v = (-1, 0, 0)$

$F = (0, +1, 0)$

$(-1, 0, 0) \times (B_x, B_y, B_z) = (0, 1, 0)$

$\Rightarrow -v_x B_z = +1 \Rightarrow B_z = +1$ B points out of page $(+z)$

2)

Coil 1:

- Switch closes $\rightarrow$ induced counter clockwise current
- long time later ($dl/dt \approx 0$) no induced current
- Switch opens $\rightarrow$ (solenoid current decays, flux into page drops) $\rightarrow$ induced clockwise current

Coil 2: exactly same pattern

1 $\rightarrow$ closing $\rightarrow$ counter clockwise

2 $\rightarrow$ steady $\rightarrow 0$

3 opening $\rightarrow$ clockwise

coil 3 $\rightarrow$ again identical

ring's radius $r = \frac{2.2 \text{ cm}}{2} = 1.1 \times 10^{-2} \text{ m}$
 area $A = \pi r^2 = \pi (1.1 \times 10^{-2})^2 \approx 3.80 \times 10^{-4} \text{ m}^2$

3.1 a) Average EMF

Initial flux $= \Phi = (2 \text{ T}) (3.8 \times 10^{-4} \text{ m}^2) = 7.6 \times 10^{-4} \text{ Wb}$

Over $\Delta t = 0.250 \text{ s}$

$\mathcal{E}_{\text{avg}} = \frac{\Delta \Phi}{\Delta t} = \frac{7.6 \times 10^{-4}}{0.250} = 3.04 \times 10^{-3} \text{ V} \quad (3.04 \text{ mV})$

b) Induced current

with $R = 0.0100 \Omega$

$I = \frac{\mathcal{E}}{R} = \frac{3.04 \times 10^{-3}}{0.0100} = 0.304 \text{ A}$

c) Power Dissipated

$P = I^2 R = (0.304)^2 (0.0100) \approx 9.25 \times 10^{-4} \text{ W}$

(0.925 mW)

4) a) Number of turns N

given $\mathcal{E}_{\text{max}} = 18 \text{ V}$, $B = 0.640 \text{ T}$, $\omega = 1875 \text{ rad/s}$

Solve for N $\frac{\mathcal{E}_{\text{max}}}{BA\omega} = \frac{18.0}{(0.640)(3 \times 10^{-4})(1875)} = \frac{18}{0.360} = 50$

Coil has 50 turns

b) period of the AC output

Angular speed $\omega = 1875 \text{ rad/s}$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{1875} \approx 3.35 \times 10^{-3} \text{ s}$

frequency $f = 1/T \approx 298.5 \text{ Hz}$

5) a) Turns ratio

$$\frac{N_P}{N_S} = \frac{V_P}{V_S} = \frac{240}{120} = 2 \Rightarrow N_P : N_S = 2 : 1$$

b) current ratio

$$\frac{I_P}{I_S} = \frac{V_S}{V_P} = \frac{120}{240} = 0.5 \Rightarrow I_P : I_S = 1 : 2$$

Secondary current is twice the primary current

6) a) Primary emf $V(t)$

$$V(t) = V_0 \sin(\omega t) \quad \text{with } V_0 = 120 \text{ V}, \omega = 240\pi \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{240\pi} = \frac{1}{120} \text{ s} \approx 8.33 \text{ ms}$$

b) Secondary (Coil 3) emf

Coil 3 has $N_3 = 1$ turn, Solenoid has $N_1 = 6$ turns

$$V_3(t) = -\frac{N_3}{N_1} V(t) = -\frac{1}{6} 120 \sin(\omega t) = -20 \sin(\omega t) \text{ V}$$

Sine wave of amplitude 20 V, 180° out of phase with $V_1(t)$

c) Zeros of $V(t)$

$$120 \sin(\omega t) = 0 \Rightarrow \sin(\omega t) = 0 \Rightarrow \omega t = n\pi, n = 0, 1, 2, \dots$$

$$t = \frac{n\pi}{\omega} = \frac{n\pi}{240\pi} = \frac{n}{240} \text{ s}$$

Thus $V(t) = 0$ at

$$t = 0, \frac{1}{240} \text{ s} \approx 4.17 \text{ ms}, \frac{2}{240} \text{ s} \approx 8.33 \text{ ms}$$
$$\frac{3}{240} \text{ s} = 12.5 \text{ ms}, \dots$$

$$7. \mathcal{E} = L \frac{\Delta I}{\Delta t}$$

$$\Delta t = \frac{L \Delta I}{\mathcal{E}} = \frac{(2.00 \times 10^{-3} \text{ H})(0.100 \text{ A})}{500 \text{ V}} = 4.0 \times 10^{-7} \text{ s}$$

8. a) Induced emf when shutting off

$$\mathcal{E} = L \frac{\Delta I}{\Delta t} = (25.0 \text{ H}) \frac{100 \text{ A}}{0.0800 \text{ s}} = 3.125 \times 10^4 \text{ V}$$

b) energy stored at 100 A

$$U = \frac{1}{2} L I^2 = \frac{1}{2} (25.0 \text{ H})(100 \text{ A})^2 = 1.25 \times 10^5 \text{ J}$$

c) Avg power to dissipate that energy in 0.0800 s

$$P_{\text{avg}} = \frac{U}{\Delta t} = \frac{1.25 \times 10^5 \text{ J}}{0.0800 \text{ s}} = 1.56 \times 10^6 \text{ W}$$

d) Example solenoid design

$$L = \mu_0 \frac{N^2 A}{l}, \quad N_0 = 4\pi \times 10^{-7} \frac{\text{H}}{\text{m}}$$

$$\text{Length } l = 0.50 \text{ m}$$

$$\text{Area } A = 1 \times 10^{-2} \text{ m}^2 \quad (\text{radius } 5.6 \text{ cm})$$

$$N = \sqrt{\frac{L l}{\mu_0 A}} = \sqrt{\frac{(25.0)(0.50)}{(4\pi \times 10^{-7})(1 \times 10^{-2})}} \approx 3.15 \times 10^4 \text{ turns}$$

e) reactance at 10 kHz

$$X_L = \omega L = 2\pi f L = 2\pi (1 \times 10^4 \text{ Hz})(25.0 \text{ H}) \approx 1.57 \times 10^6 \Omega$$

9) a) Inductance

$$L = \tau R = (30 \times 10^{-9} \text{ s}) (5 \times 10^6 \Omega) = 0.100 \text{ H}$$

b) resistance for $\tau = 1.0 \text{ ns}$

$$R = \frac{L}{\tau} = \frac{0.100 \text{ H}}{1 \times 10^{-9} \text{ s}} = 1 \times 10^8 \Omega = 100 \text{ M}\Omega$$

c) Fraction of I_0 after 3.0 ns

$$\frac{i(3 \text{ ns})}{I_0} = e^{-3.0 \text{ ns} / 20.0 \text{ ns}} = e^{-0.15} \approx 0.861$$

about 86.1% of the original current remains 3 ns after shut off.

10) a)

$$v_{in}(t) - i(t)R - v_{out}(t) = 0$$

$$b) v_{out}(t) = \frac{1}{C} \int i(t) dt.$$

$$c) i(t) = C \frac{dv_{out}}{dt}$$

$$v_{in}(t) - RC \frac{dv_{out}}{dt} - v_{out}(t) = 0 \Rightarrow RC \frac{dv_{out}}{dt} + v_{out}(t) = v_{in}(t)$$

$$RC(j\omega)V_{out} + V_{out} = V_{in} \Rightarrow \frac{V_{out}}{V_{in}} = \frac{1}{1 + j\omega RC} = H(j\omega)$$

$$d) |H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\text{for } f = 100 (2\pi RC)^{-1} \text{ we have: } \omega = 2\pi f = \frac{100}{RC} \text{ so } \omega RC = 100$$

$$|H| = \frac{1}{\sqrt{1 + 100^2}} \approx \frac{1}{100.005} \approx 0.010$$

For $f = 0.1 (2\pi RC)^{-1}$ $\omega RC = 0.1$ (a) (p)

$$|H| = \frac{1}{\sqrt{1+0.1^2}} = \frac{1}{\sqrt{1.01}} \approx 0.995$$

11. a) $f_0 = \frac{1}{2\pi\sqrt{LC}}$ $L = 10 \text{ mH} = 1 \times 10^{-2} \text{ H}$, $C = 1 \text{ pF} = 1 \times 10^{-6} \text{ F}$

$$f_0 = \frac{1}{2\pi\sqrt{(1 \times 10^{-2})(1 \times 10^{-6})}} = \frac{1}{2\pi(1 \times 10^{-4})} \approx 1.59 \times 10^3 \text{ Hz} = 1591.6 \text{ Hz}$$

b) $Z(\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right)$ where $R = 0.1 \text{ k}\Omega = 100 \Omega$

At $f = 0.1 f_0$

$$\omega_0 = 2\pi f_0 = 2\pi(1591.6) \approx 1 \times 10^4 \text{ rad/s}$$

$$\omega = 0.1\omega_0 = 1 \times 10^3 \text{ rad/s}$$

$$X_L = \omega L = (1 \times 10^3)(1 \times 10^{-2}) = 10 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(1 \times 10^3)(1 \times 10^{-6})} = 1000 \Omega$$

total impedance: $Z = 100 - j(1000 - 10) = 100 - j990 [\Omega]$

At $f = 10 f_0$

$$\omega = 10\omega_0 = 1 \times 10^5 \text{ rad/s}$$

$$X_L = (1 \times 10^5)(1 \times 10^{-2}) = 1000 \Omega$$

$$X_C = \frac{1}{(1 \times 10^5)(1 \times 10^{-6})} = 10 \Omega$$

total impedance = $100 + j(1000 - 10) = 100 + j990 [\Omega]$

12. a) I_{rms}

$$I_{rms} = \frac{V_{rms}}{|Z|} = \frac{120}{995} \approx 0.1207 \text{ A}$$

because $|Z|$ is the same at $0.1 f_0$ and $10 f_0$

$$I_{rms} \approx 0.121 \text{ A at both } 0.1 f_0 \text{ and } 10 f_0$$

b) P_{rms}

$$P_{rms} = V_{rms} I_{rms} \cos \phi = I_{rms}^2 R$$

$$\text{since } \cos \phi = \frac{R}{|Z|} = \frac{100}{995} \approx 0.1005$$

$$P_{rms} = 120 \times 0.1207 \times 0.1005 \approx 1.45 \text{ W}$$

at both frequencies $P_{rms} \approx 1.45 \text{ W}$

13. Carrier at

$$f_c = 1.4 \text{ MHz}$$

Upper sideband at

$$f_c + f_m = 1.4 \text{ MHz} + 0.010 \text{ MHz} = 1.410 \text{ MHz}$$

Lower sideband at

$$f_c - f_m = 1.4 \text{ MHz} - 0.010 \text{ MHz} = 1.390 \text{ MHz}$$

Unit 5:

$$1. a) I_d = \frac{\Delta Q}{\Delta t} = \frac{200 \times 10^{-9} \text{ C}}{2 \times 10^{-6} \text{ s}} = 0.100 \text{ A}$$

$$\oint \mathbf{B} \cdot d\mathbf{I} = \mu_0 I_{enc} \Rightarrow B(2\pi r) = \mu_0 I_d$$

$$B = \frac{\mu_0 I_d}{2\pi r} = \frac{(4\pi \times 10^{-7})(0.100)}{2\pi(0.01)} = 2 \times 10^{-6} \text{ T}$$

b) displacement current

$$I_d = \frac{dQ}{dt} = 0.10 \text{ A}$$

$$2. a) 2d = c \Delta t \Rightarrow d = \frac{c \Delta t}{2} = \frac{(3 \times 10^8 \text{ m/s})(10 \times 10^{-6} \text{ s})}{2} = 1.5 \times 10^3 \text{ m}$$

aircraft is approximately 1.5 km above ice

$$b) K = \left| \frac{n_2 - n_1}{n_2 + n_1} \right|^2 \quad n_{air} \approx 1.0 \quad n_{ice} \approx \sqrt{\epsilon_r} \approx \sqrt{2.17} \approx 1.47$$

$$K = \left| \frac{1.47 - 1.00}{1.47 + 1.00} \right|^2 = \left(\frac{0.47}{2.47} \right)^2 \approx (0.190)^2 \approx 0.036$$

about 3.6% of incident power is reflected by ice surface

3. a) $A_1 = 10 \text{ cm}^2 = 10 \times 10^{-4} \text{ m}^2 = 1 \times 10^{-3} \text{ m}^2$

with $P = 10 \times 10^3 \text{ W}$

$$I_1 = \frac{P}{A_1} = \frac{1 \times 10^3 \text{ W}}{1 \times 10^{-3} \text{ m}^2} = 1 \times 10^6 \text{ W/m}^2$$

b) because intensity falls off as $1/r^2$, $r = 2 \text{ m}$

$$I_2 = I_1 \left(\frac{1}{2^2} \right) = 1 \times 10^6 \text{ W/m}^2 \times \frac{1}{4} = 2.5 \times 10^5 \text{ W/m}^2$$

4. a) proof that $\theta_3 = \theta_1$ when $n_1 = n_3$

1st interface (n_1 to n_2) $n_1 \sin \theta_1 = n_2 \sin \theta_2$

2nd interface (n_2 to n_3) $n_2 \sin \theta_2 = n_3 \sin \theta_3$

if $n_1 = n_3$ $n_1 \sin \theta_1 = n_3 \sin \theta_3 \Rightarrow \sin \theta_1 = \sin \theta_3 \Rightarrow \theta_3 = \theta_1$

b) $M = 1 + \frac{D}{F}$ here $F = 12.0 \text{ cm}$

$$M = 1 + \frac{25}{12} \approx 1 + 2.08 = 3.08$$

doctor sees details about 3.1 times larger than naked eye

5. $\frac{I}{I_0} = e^{-Nt}$ $\rho = N\sigma$ $\rho = 11.34 \text{ g/cm}^3$ $M = 207.2 \text{ g/mol}$

$$N = \frac{\rho}{M} N_A = \frac{11.34 \text{ g/cm}^3}{207.2 \text{ g/mol}} (6.022 \times 10^{23} \text{ mol}^{-1}) \approx 3.3 \times 10^{22} \text{ atoms/cm}^3$$

$$\sigma = 200 \text{ barns} = 200 \times 10^{-24} \text{ cm}^2 = 2.0 \times 10^{-22} \text{ cm}^2$$

$$\mu = N\sigma = (3.3 \times 10^{22} \text{ cm}^{-3}) (2 \times 10^{-22} \text{ cm}^2) = 6.6 \text{ cm}^{-1}$$

$$\frac{I}{I_0} = e^{-\mu x} = e^{-6.6 \times 1} \approx 1.36 \times 10^{-3}$$

about 0.14% of the 10 keV x-rays get through a 1 cm thick lead vest

6. a) $E = 250 \text{ mJ} = 0.250 \text{ J}$, $m = 60 \text{ kg}$

$$D = \frac{E}{m} = \frac{0.250 \text{ J}}{60 \text{ kg}} = 4.17 \times 10^{-3} \text{ Gy}$$

$$D_{\text{rad}} = 4.17 \times 10^{-3} \text{ Gy} \times 100 \frac{\text{rad}}{\text{Gy}} \approx 0.417 \text{ rad}$$

b) $E = 0.250 \text{ J}$, $m = 2 \text{ kg}$

$$D = \frac{0.250}{2} = 0.125 \text{ Gy} \Rightarrow D_{\text{rad}} = 0.125 \times 100 = 12.5 \text{ rad}$$

c) x- / y-rays, 1 Gy $\rightarrow$ 1 Sv

$$D_{\text{SV}} = 0.125 \text{ Gy} = 0.125 \text{ Sv} = 125 \text{ mSv}$$

I do not think it would cause major radiation sickness but perhaps 125 mSv is enough to warrant minimizing exposure